INCH-POUND

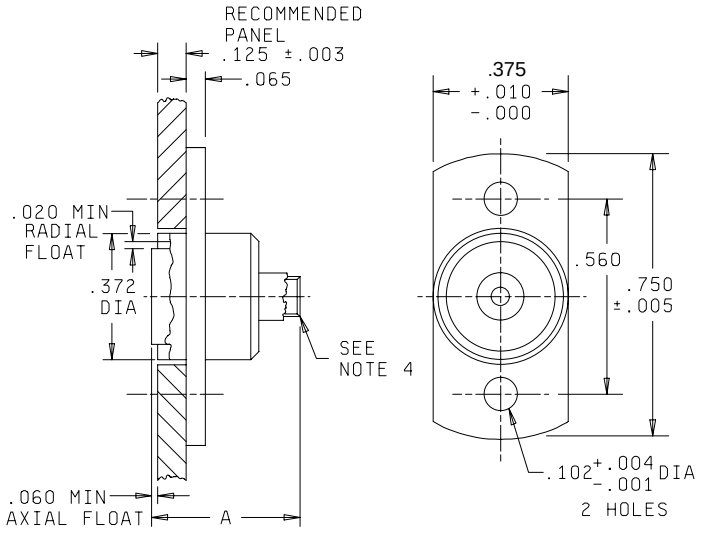
MIL-PRF-31031/1A
3 November 1998
SUPERSEDING
MIL-C-31031/1
19 January 1995

PERFORMANCE SPECIFICATION

# CONNECTOR, RECEPTACLE, ELECTRICAL, RADIO FREQUENCY, SERIES BMA, FLANGE MOUNT, FOR FLEXIBLE CABLE, SOCKET CONTACT, HIGH RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-31031.



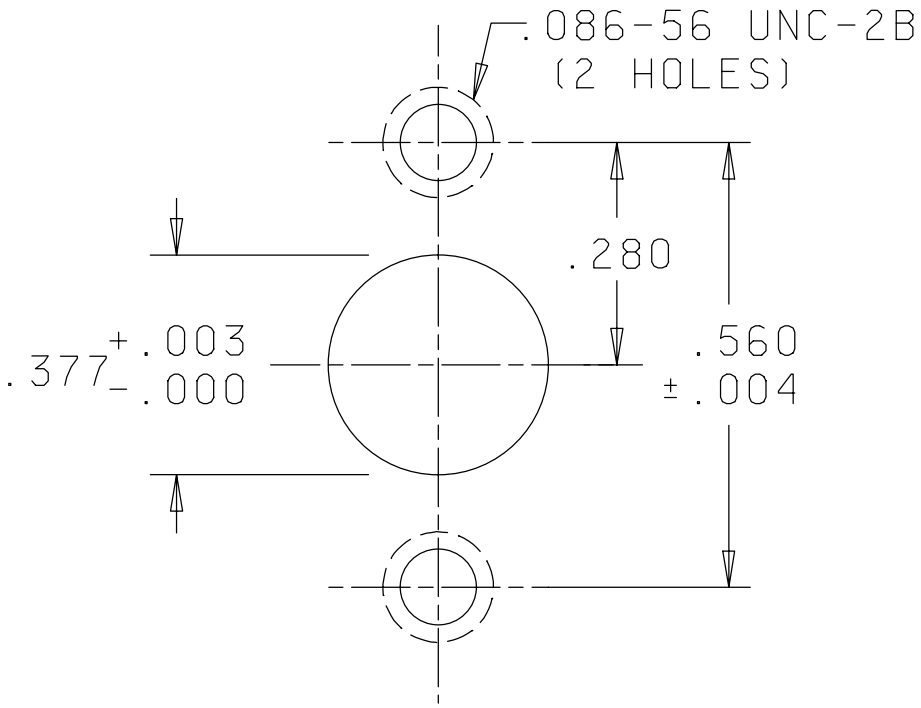

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .000 | 0.00 | .010 | 0.25 | .125 | 3.18 |
| .001 | 0.03 | .020 | 0.51 | .372 | 9.45 |
| .003 | 0.08 | .060 | 1.52 | .375 | 9.53 |
| .004 | 0.10 | .065 | 1.65 | .560 | 14.22 |
| .005 | 0.13 | .102 | 2.59 | .750 | 19.05 |

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, all tolerances are ± .005 inch.
4. See table I for cable accommodation.
5. For dimension A, see table I.
6. Dimension A defines the overall length of the connector when assembled to the cable.

FIGURE 1. General configuration.


DISTRIBUTION STATEMENT A. Approved for public release distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T02:17Z
Check the source to verify that this is the current version before use.




RECOMMENDED MOUNTING HOLE

| Inches | mm |
|---|---|
| .000 | 0.00 |
| .003 | 0.08 |
| .004 | 0.10 |
| .086 | 2.18 |
| .280 | 7.11 |
| .377 | 9.58 |
| .560 | 14.22 |

Spring compression force and travel: The face of the push-on interface shall have a total travel capability of .060 inch toward the connector retainer clip. Force versus displacement shall be as follows:

| Displacement Inches | Force pounds |
|---|---|
| .010 | 3.0 ± .5 |
| .050 | 3.7 ± .5 |
| .060 | 3.8 ± .5 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Recommended mounting hole.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T02:17Z
Check the source to verify that this is the current version before use.

TABLE I. PIN's cross-reference and dimensions.

| PIN 1/ M31031/01 | Applicable cable and group 2/ M17/ | Dimensions | Inches (millimeters) maximum |
|---|---|---|---|
| Category A - Field serviceable (no special tools required) | | | |
| A**01 | 113-RG316 group II | A | 1.4 (35.56 mm) |
| A**02 | 128-RG400 group VI | A | 1.4 (35.56 mm) |

1/ These connectors have captivated center contacts.

2/ The latest version of the cable is applicable.

TABLE II. Cable retention force.

| Cable dielectric outer diameter | Pounds (minimum) | |
|---|---|---|
| | Single braid | Double braid |
| Inches (maximum) | | |
| .036 | 10 | N/A |
| .067 | 20 | N/A |
| .110 | 30 | N/A |
| .122 | 40 | 45 |

TABLE III. Group qualification. 1/ 2/ 3/ 4/

| Group | Submission and qualification of any of the following connectors | Qualifies the following connectors |
|---|---|---|
| I | M31031/01A**02 | M31031/01A**01<br>M31031/01A**02 |

* Options as defined in the PIN construction.

1/ Individual connectors other than those listed are self qualifying.

2/ Qualification of connectors qualifies connectors of the same body materials only.

3/ Qualification of connectors qualifies connectors of the same environmental construction only.

4/ Changes in plating (body or center contact) shall require the corrosion and contact resistance tests be performed.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T02:17Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA

Nominal impedance: 50 ohms ± 1 ohm.

Frequency range: 0 to 18 GHz (frequency limit is dependent on cable used).

Voltage rating:
Cable group II: 250 V rms at sea level: 65 V rms at 70,000 feet.
Cable groupVI: 335 V rms at sea level: 85 V rms at 70,000 feet.

Temperature rating: -65°C to +125°C.

REQUIREMENTS: The following tests assure connector integrity within typical operating conditions and applications. Alternate methods shall be made known to the qualifying authority prior to performance of the test. The test methods described in the general specification and herein are proven methods and shall be the refree method in case of dispute.

Dimensions and configurations: See figures 1 and 2 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force: 3 pounds maximum.
Torque: Not applicable.
Radial misalignment: ± .020 inch minimum.
Axial misalignment: .060 inch minimum.
Coupling proof torque: Not applicable.

Mating characteristics:

Reference MIL-STD-348 for dimensions.

Hermetic seal: Not applicable.

Permeability: Applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 3003 of MIL-STD- 1344. 5,000 megohms minimum.

Center contact retention: 6 pounds minimum axial force.

Radial torque: Not applicable.

Corrosion (salt spray): Method 1001 of MIL-STD-1344, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 18 GHz or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower:

Cable group II: 1.15 + .02 F (GHz).
Cable group VI: 1.15 + .01 F (GHz).

Connector durability: 500 cycles minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum:

| | Initial | After environment |
|---|---|---|
| Center contact | 3.0 | 4.0 |
| Outer contact | 2.0 | Not applicable |
| Braid to body | 0.5 | Not applicable |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T02:17Z
Check the source to verify that this is the current version before use.

Dielectric withstanding voltage: Method 3001 of MIL-STD-1344.

Cable group II: 750 volts rms at 60 Hz at sea level.
Cable group VI: 1,000 volts rms at 60 Hz at sea level.

Vibration high frequency: Method 2005 of MIL-STD-1344, test condition III. No discontinuity permitted. 1/

Shock: Method 2004 of MIL-STD-1344, test condition F. No discontinuity permitted. 1/

Temperature cycling: Method 1003 of MIL-STD-1344, test condition A, except test high temperature shall be +200°C for connectors using +200°C cables (see table I). Low test temperature shall be -65°C.

Moisture resistance: Method 1002 of MIL-STD-1344, test condition B. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level (at 70,000 feet): 1/

Cable group II: 190 volts rms minimum.
Cable group VI: 250 volts rms minimum.

RF high potential withstanding voltage: 1/

Voltage and frequency (at a frequency from 5 to 7.5 MHz):

Leakage current: Not applicable.

Cable group II: 500 volts rms minimum.
Cable group VI: 670 volts rms minimum.

Cable retention force: See table II.

Coupling mechanism retention force: Not applicable.

Shielding effectiveness: Method 3008 of MIL-STD-1344: -85 dB minimum. 1/

RF transmission loss (dB maximum) 1/: $.03\sqrt{\text{frequency GHz}}$. Test at a frequency between 9 and 12.4 GHz.

Porosity: Applicable.

Thermal vacuum outgassing: Applicable to space grade connectors only.

Flammability: Applicable to space grade connectors only.

Odor: Applicable to space grade connectors only.

Toxicity (offgassing): Applicable to space grade connectors only.

Safety wire hole pullout: Not applicable.

Part or Identifying Number (PIN): M31031/01 (see table I).

Group qualification: See table III.

1/ This test needs to be performed only during initial qualification, so long as the qualifying design does not change.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T02:17Z
Check the source to verify that this is the current version before use.

NOTES:

1. If a connector manufacturer produces a connector that meets all the requirements for two or more connector PIN's (within the same series and of the same environmental grade), the manufacturer may receive qualification approval for two or more connectors PIN's qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own PIN. For group qualification, the connectors must be of the similar design.

2. For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right hand column. The part does not necessarily have to be the part initially qualified.

Supersession: This specification supersedes DESC drawing 85073 when a QPL source becomes available.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project 5935-4120-01)

Review activities:
Army - AT, AV, EA, MI
Navy - AS, MC, OS, SH
Air Force - 17, 19, 99



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T02:17Z
Check the source to verify that this is the current version before use.